Blockchain Experiment 10

Aim: Explore the Cryptocurrency Landscape

Theory:

This experiment explores the cryptocurrency landscape by combining real-world market data analysis with blockchain technology research. The focus asset for this analysis is Chainlink (LINK), and data was collected from three reputable sources as per the lab requirement:

1. **CoinGecko API** – Provided historical price data, market capitalization, and trading volumes in a structured JSON format, enabling direct processing in Python.
2. **CoinMarketCap** – Offered real-time and historical price charts, circulating supply, and market rank for visual cross-verification.
3. **Crypto.com** – Supplied quick-access market statistics, simplified price trends, and cryptocurrency news updates.

Cryptocurrency Landscape

1. Definition – Cryptocurrencies are decentralized digital assets built on blockchain networks, enabling secure, transparent, and immutable transactions without central authority.
2. Market Diversity – Thousands of cryptocurrencies exist, each serving unique roles:
    - Store of Value – Bitcoin, Litecoin.
    - Utility Tokens – Chainlink, Ethereum.
    - Stablecoins – USDT, USDC.
    - Governance Tokens – UNI, AAVE.
3. Volatility – Prices are highly sensitive to market sentiment, technological developments, and regulatory updates.
4. Data Analysis Importance – Tracking historical trends, market capitalization, and trading volume is essential for understanding asset performance.
5. Advancements – Growing use of DeFi platforms, NFT ecosystems, layer-2 scaling, and cross-chain protocols.

Data Collection Process

The program used the CoinGecko API to automate historical data fetching for Chainlink. The retrieved data was processed with Pandas to compute:

- Daily Returns – Percentage change between consecutive days to measure volatility.
- Moving Averages (MA7, MA30) – Short-term and long-term trend indicators.
- Cumulative Returns – Growth factor over time.
- Rolling Volatility – Short-term risk measurement based on daily return fluctuations.

Visualization & Dashboard

Using Matplotlib, a six-panel dashboard was created to display:

1. Price Trend – Daily price movement over the selected period.
2. Price vs Volume – Correlation between trading activity and price changes.
3. Daily Returns Distribution – Statistical spread of daily gains/losses.
4. Moving Averages – Technical analysis trend lines for price movement.
5. Cumulative Returns – Overall growth performance.
6. Rolling Volatility – Market risk trends over time.

Technological Analysis

Beyond market data, the study covered advancements in blockchain technology, including:

- Consensus Mechanisms – PoW (Proof of Work), PoS (Proof of Stake), and hybrid systems.

- Scalability Solutions – Layer-2 protocols, sharding, and sidechains to increase throughput.
- Interoperability Proto
- cols – Chainlink's CCIP, Polkadot, and Cosmos enabling cross-chain data exchange.Privacy Features – Zero-knowledge proofs (zk-SNARKs) and ring signatures to ensure transaction confidentiality.

Chainlink's roadmap highlights staking implementation, advanced oracle features, and enhanced cross-chain capabilities, aligning with industry trends toward decentralization, interoperability, and security.

Market Trends & Adoption

- Growing Merchant Acceptance – More businesses accepting cryptocurrency payments.
- Wallet Growth – Increasing downloads and installations of crypto wallets.
- Network Activity – Higher transaction volumes and active addresses, signaling strong adoption.
- Integration with Traditional Finance – Partnerships between crypto platforms and fintech/payment providers.
- Regulatory Developments – Countries exploring frameworks for safe cryptocurrency adoption.

Utilizing Api for analysis and visualization :

Code:

```
import requests
import matplotlib.pyplot as plt
import pandas as pd

# ---------- Fetch Data ----------
def get_historical_data(coin_id, days='180', currency='usd'):
    url = f'https://api.coingecko.com/api/v3/coins/{coin_id}/market_chart'
    params = {'vs_currency': currency, 'days': days}
    response = requests.get(url, params=params)
    return response.json()

# ---------- Dashboard Function ----------
def chainlink_dashboard(days='180'):
    coin_id = 'chainlink'
    data = get_historical_data(coin_id, days)

    # Create DataFrame
    prices_df = pd.DataFrame(data['prices'], columns=['timestamp', 'price'])
    volumes_df = pd.DataFrame(data['total_volumes'], columns=['timestamp', 'volume'])

    prices_df['timestamp'] = pd.to_datetime(prices_df['timestamp'], unit='ms')
    volumes_df['timestamp'] = pd.to_datetime(volumes_df['timestamp'], unit='ms')

    df = pd.merge(prices_df, volumes_df, on='timestamp')
    df.set_index('timestamp', inplace=True)

    # Calculate extra metrics
    df['returns'] = df['price'].pct_change()
    df['MA7'] = df['price'].rolling(window=7).mean()
    df['MA30'] = df['price'].rolling(window=30).mean()
    df['cumulative'] = (1 + df['returns']).cumprod()
    df['volatility'] = df['returns'].rolling(window=7).std() * 100

    # ---------- Plot Dashboard ----------
    fig, axes = plt.subplots(3, 2, figsize=(16, 12))
    fig.suptitle(f'{coin_id.capitalize()} Dashboard (Last {days} Days)', fontsize=16, fontweight='bold')
```

```
    # 1. Price Trend
    axes[0, 0].plot(df.index, df['price'], color='blue')
    axes[0, 0].set_title('Price Trend')
    axes[0, 0].set_ylabel('Price (USD)')
    axes[0, 0].set_xlabel('Year-Month') # Add xlabel
    axes[0, 0].grid(True)

    # 2. Price vs Volume
    axes[0, 1].plot(df.index, df['price'], color='blue', label='Price')
    axes[0, 1].bar(df.index, df['volume'], color='orange', alpha=0.3, label='Volume')
    axes[0, 1].set_title('Price vs Volume')
    axes[0, 1].set_xlabel('Year-Month') # Add xlabel
    axes[0, 1].set_ylabel('Price (USD) / Volume') # Add ylabel
    axes[0, 1].legend()
    axes[0, 1].grid(True)

    # 3. Daily Returns Distribution
    axes[1, 0].hist(df['returns'].dropna(), bins=50, alpha=0.7, color='purple')
    axes[1, 0].set_title('Daily Returns Distribution')
    axes[1, 0].set_xlabel('Daily Return')
    axes[1, 0].set_ylabel('Frequency')
    axes[1, 0].grid(True)

    # 4. Moving Averages
    axes[1, 1].plot(df.index, df['price'], color='lightgray', label='Price')
    axes[1, 1].plot(df.index, df['MA7'], color='blue', label='7-day MA')
    axes[1, 1].plot(df.index, df['MA30'], color='red', label='30-day MA')
    axes[1, 1].set_title('Moving Averages')
    axes[1, 1].set_xlabel('Year-Month')
    axes[1, 1].set_ylabel('Price (USD)')
    axes[1, 1].legend()
    axes[1, 1].grid(True)

    # 5. Cumulative Returns
    axes[2, 0].plot(df.index, df['cumulative'], color='green')
    axes[2, 0].set_title('Cumulative Returns')
    axes[2, 0].set_ylabel('Growth Factor')
    axes[2, 0].set_xlabel('Year-Month')
    axes[2, 0].grid(True)

    # 6. Volatility
    axes[2, 1].plot(df.index, df['volatility'], color='orange')
    axes[2, 1].set_title('Rolling Volatility (7-Day)')
    axes[2, 1].set_ylabel('Volatility (%)')
    axes[2, 1].set_xlabel('Year-Month')
    axes[2, 1].grid(True)

    plt.tight_layout(rect=[0, 0, 1, 0.96])
    plt.subplots_adjust(hspace=0.5, wspace=0.3) # Adjust vertical and horizontal spacing
    plt.show()
# ---------- Run Dashboard ----------
chainlink_dashboard('180')
```

Output:

coingecko.com/en/all-cryptocurrencies

#	Coin	Price	1h	24h	7d	30d	24h Volume	Circulating Supply	Total Supply	Market Ca
7	USDC USDC	$0.9997	0.0%	0.0%	0.0%	0.0%	$15,401,345,704	67,082,929,588	67.12B	$67,063,626,61
8	Lido Staked Ether STETH	$4,756.42	0.5%	2.8%	30.3%	60.2%	$177,982,377	8,841,808	8.84M	$42,052,347,08
9	Dogecoin DOGE	$0.2529	0.1%	5.9%	24.1%	31.5%	$3,908,067,633	150,502,056,384	150.51B	$38,062,557,10
10	Cardano ADA	$0.9815	0.6%	14.6%	33.3%	35.7%	$2,950,310,906	36,362,934,087	45B	$35,695,074,50
11	TRON TRX	$0.3644	0.3%	2.8%	7.7%	21.1%	$1,432,964,578	94,688,682,529	94.69B	$34,499,424,27
12	Wrapped stETH WSTETH	$5,761.79	0.5%	2.8%	30.5%	60.5%	$96,443,244	3,198,384	3.2M	$18,427,655,86
13	Chainlink LINK	$24.28	1.0%	0.7%	45.9%	56.6%	$1,951,447,602	678,099,970	1B	$16,420,608,81
14	Hyperliquid HYPE	$48.54	1.6%	8.7%	27.9%	1.6%	$530,339,658	333,772,999	999.84M	$16,199,716,67
15	Wrapped Bitcoin WBTC	$123,432	0.1%	3.3%	7.7%	4.6%	$330,809,526	127,038	127037.58	$15,68
16	Wrapped Beacon ETH WBETH	$5,128.38	0.5%	2.6%	30.5%	60.7%	$52,896,879	2,964,154	2.96M	$15,174,319,71

https://www.coingecko.com/en/coins/chainlink

Cryptocurrencies > Chainlink Price

Chainlink LINK Price #13

$23.47 9.5% (24h)

0.0001967 BTC 8.9%
0.005079 ETH 1.6%

$21.07 24h Range $24.52

Add to Portfolio • 585,889 added

Market Cap	$15,922,260,596
Fully Diluted Valuation	$23,480,697,964
24 Hour Trading Vol	$2,666,194,693
Circulating Supply	678,099,970
Total Supply	1,000,000,000
Max Supply	1,000,000,000

Buy / Sell Wallet Earn Crypto

Info

Contract 0x514...f986ca

Overview Markets News Similar Coins Historical Data

Price Market Cap Live Chart 24h 7d 1m 3m 1y Max LOG

$25.00 $24.00 $23.00 $22.00 $21.00 $20.00

coingecko

15:00 18:00 21:00 13. Aug 03:00 06:00 09:00 12:00

2018 2020 2022 2024

Compare with: BTC ETH

Need more data? Explore our API

https://coinmarketcap.com/currencies/chainlink/

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Chainlink LINK #11 1M

$23.39 8.79% (1d)

Why is LINK's price up today?

Market cap $15.86B 8.8%	
Volume (24h) $2.6B 123.33%	FDV $23.39B
Vol/Mkt Cap (24h) 16.14%	Total supply 1B LINK
Max. supply --	Circulating supply 678.09M LINK

Profile score 61%

Website Website Whitepaper

Socials

C... 0x5149...f986ca

Chart Markets News Yield About Buy LINK

Price Market cap 1D 7D 1M 1Y All LOG

24.00 23.39 23.00 22.00 21.5847 21.00 20.00 USD

CoinMarketCap

12:00 PM 4:00 PM 8:00 PM 13 Aug 4:00 AM 8:00 AM 12:00 PM

15 Chainlink integrates...

Chainl... 819.5K Followers + Follow

Community sentiment 818.4K votes 1/2

86% 14%

Bullish Bearish

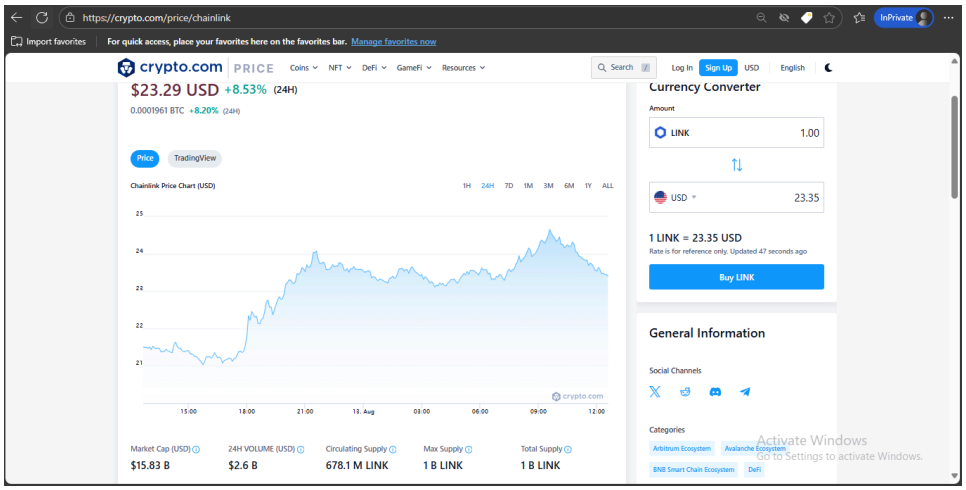
Top Latest

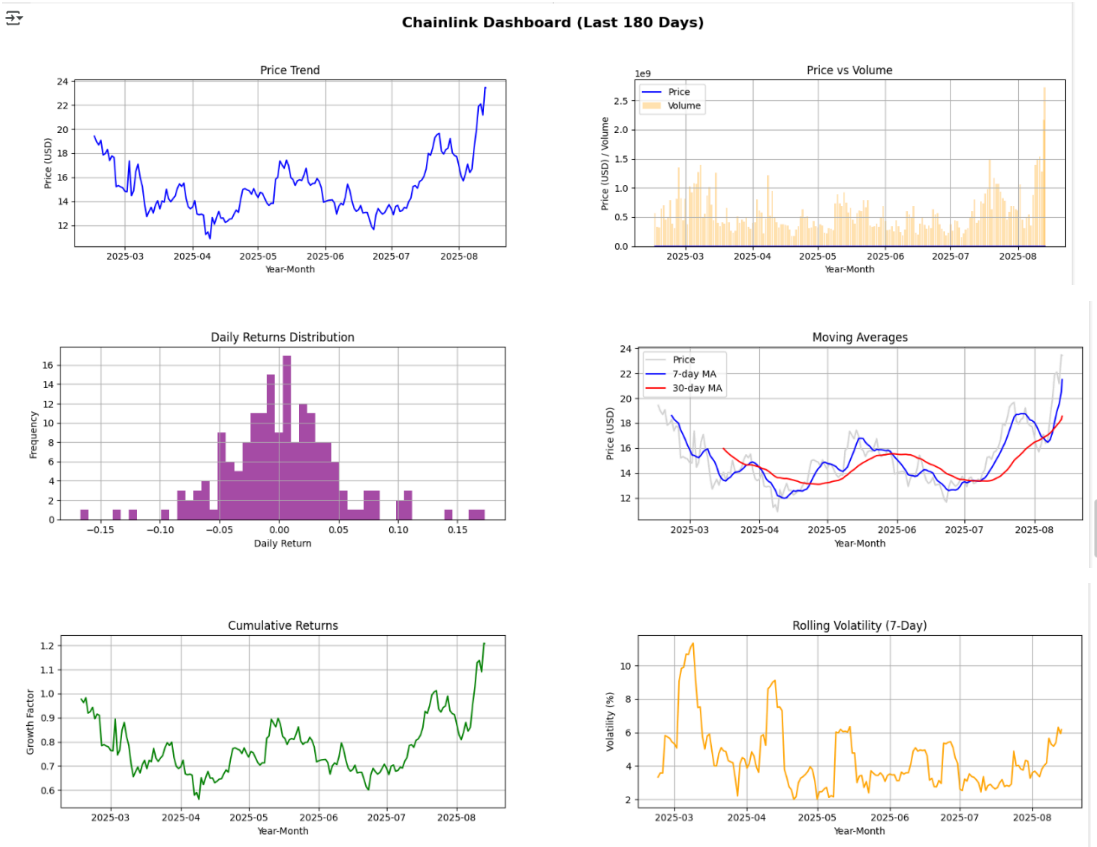
Adex_crypt · 18 hours + Follow

$LINK is showing strong long-term potential.

Price has broken decisively above the $20 resistance, signaling a bullish ma ... Read all

$LINK How do you feel to... Post





Conclusion:

By combining automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, this experiment presents a comprehensive and holistic view of the cryptocurrency ecosystem. It goes beyond isolated technical or financial analysis by integrating multiple perspectives, allowing for a deeper understanding of how different components of the ecosystem interact with one another. This includes not only price movements and market sentiment but also the underlying technological innovations that drive these digital assets.

Through the study of major platforms like Bitcoin and Ethereum, the experiment highlights how blockchain infrastructure, consensus mechanisms, and smart contract capabilities influence real-world market behavior. Visual trend analysis further enhances this understanding by making complex data more accessible, enabling the identification of patterns, cycles, and anomalies that may not be immediately apparent through raw data alone.